

Paradise Lost? Using Synthetic Historical Controls to Quantify Hawaii’s Mandatory Quarantine on Tourism

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Summary This paper uses a novel causal time series method to estimate the causal impact of Hawaii’s quarantine policy to combat the spread of COVID-19 on Hawaii’s tourism industry.

Keywords: *Synthetic Controls, Policy Evaluation, COVID-19*

1. INTRODUCTION

Unlike other regions where economic contraction from COVID-19 emerged organically (through fear, voluntary distancing, or inconsistent mandates) Hawaii’s economic/tourism collapse was the direct and intended consequence of state policy. On March 26 2020, the government took the extraordinary step of effectively sealing the state’s borders, mandating a 14-day quarantine for all arrivals regardless of origin. In doing so, Hawaii enacted one of the few truly conscious economic shutdowns during the pandemic: a policy designed not just to mitigate risk, but to actively suppress travel and commerce. [Ivanova \(2020\)](#) documents that a third of Hawaii’s labor force was unemployed within a month following the border closure, suggesting immediate economic impact.

As argued by [Callaway and Li \(2023\)](#) and [Bilgel \(2022\)](#), conventional comparative case study methods break down when all units are treated. Both difference-in-differences (DiD) and synthetic control methods (SCM) depend on comparing a treated unit to a set of untreated units, under the assumption that, absent treatment, their outcomes would have followed either parallel trends or would be a convex combination of control units. In the COVID-19 context, this assumption is problematic. All U.S. states experienced pandemic-induced economic shocks, even if their formal policies varied. Thus, no state/island nation can be considered a credible untreated control for Hawaii.

Despite this, researchers have applied SCM to study the effects of non-pharmaceutical interventions (NPIs) on health and economic outcomes during the pandemic [e.g., [Yang \(2021\)](#); [Friedson et al. \(2021\)](#); [Gibson and Sun \(2023\)](#); [Breidenbach and Mitze \(2021\)](#); [Bulle et al. \(2022\)](#)]. Others use regression based panel data approaches [Ke and Hsiao \(2022, 2023\)](#). These papers contribute valuable insights, but they all operate within the comparative case study framework where geographic control units exist, where certain cities or counties/regions would mandate masks and others would not. To overcome the necessity of an untreated donor pool, I use the *Synthetic Historical Control* (SHC) method, following the framework proposed by [Chen et al. \(2024\)](#). This method does not rely on geographical controls, instead relying only on the treated unit’s history.

This paper contributes to the literature on the broader economic consequences of lockdown-style policies, beyond their direct effects on infection and mortality rates.

Understanding the economic tradeoffs of extreme containment measures can help inform future policy, especially in tourism-dependent economies. Moreover, given that most existing work has focused on interventions in mainland states [e.g., [Friedson et al. \(2021\)](#); [Morgan et al. \(2025\)](#); [Gius \(2024\)](#); [Cohn et al. \(2022\)](#); [Dave et al. \(2021, 2022\)](#)], the Hawaiian case offers a unique and underexplored setting. Methodologically, this is the first applied paper to implement the SHC method. In doing so, it demonstrates the SHC as a tool for policy evaluation when suitable donor units are unavailable.

The remainder of the paper is structured as follows. Section 2 introduces the empirical methodology. Section 3 describes the data. Section 4 presents the main findings. Section 5 concludes and discusses limitations.

2. METHODOLOGY

We know what happened given that Hawaii mandated two-week quarantines. We do not know what would have happened had they not done this. Given its novelty and radical deviation from SCM, this section summarizes the SHC method by [Chen et al. \(2024\)](#). For a comprehensive treatment of the method, including the theoretical proofs and asymptotic properties, readers are referred to [Chen et al. \(2024\)](#).

Notation Let $\mathbb{R}$ denote the set of real numbers, and let $\|\cdot\|_1$ denote the usual ℓ_1 vector norm. Throughout, I denote sets using calligraphic letters (e.g., $\mathcal{T}, \mathcal{S}, \mathcal{N}$) for clarity. Scalars are represented using plain lowercase letters (e.g., h, t, n, m), and matrices are denoted by bold uppercase letters (e.g., $\mathbf{X}, \mathbf{W}, \mathbf{Y}$). Given a vector $\mathbf{x} = (x_1, x_2, \dots, x_T)^\top \in \mathbb{R}^T$ and an index set $\mathcal{I} \subseteq \{1, \dots, T\}$, I write $\mathbf{x}_{\mathcal{I}} := (x_t)_{t \in \mathcal{I}} \in \mathbb{R}^{|\mathcal{I}|}$ to denote the subvector of $\mathbf{x}$ corresponding to indices in $\mathcal{I}$, preserving their original order.

Let $\mathcal{T} = \{1, 2, \dots, T\}$ index time at a monthly frequency. Define the pre-treatment period as $\mathcal{T}_1 := \{t \in \mathcal{T} : t \leq T_0\}$ and the post-treatment period as $\mathcal{T}_2 := \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$. Let $m \in \mathbb{N}$ denote the evaluation window length, i.e., the number of periods used to construct the SHC match. Define the evaluation period as the final m months of the pre-treatment period $\mathcal{T}_{\text{eval}} := \{T_0 - m + 1, \dots, T_0\} \subset \mathcal{T}_1$.

Let $\mathbf{y} = (y_1, y_2, \dots, y_T)^\top \in \mathbb{R}^T$ denote the observed outcome vector for the treated unit. Define the pre-treatment outcome vector $\mathbf{y}_{\text{pre}} := \mathbf{y}_{\mathcal{T}_1} \in \mathbb{R}^{T_0}$, the evaluation subvector $\mathbf{y}_{\text{eval}} := \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector $\mathbf{y}_{\text{post}} := \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$.

The evaluation subvector $\mathbf{y}_{\text{eval}}$ is the target pattern to be reconstructed using convex combinations of earlier segments from the pre-treatment period. To reconstruct the evaluation window, we compare it against earlier segments of the treated unit's own pre-treatment trajectory.

Define the donor matrix $\tilde{\mathbf{Y}}_{\text{pre}} \in \mathbb{R}^{m \times N}$ as a collection of N overlapping length- m subvectors extracted from the smoothed pre-treatment series. Each column is a contiguous subvector of the form $\tilde{\mathbf{y}}_{[i, i+m-1]}$, with i ranging over eligible start points in $\mathcal{T}_1$ that precede the evaluation window. The precise construction is given in Section~??.

2.1. Synthetic Historical Control Method

Potential Outcomes Following the Rubin potential outcomes framework, let y_t^1 and y_t^0 denote the potential outcomes for the treated unit at time $t \in \mathcal{T}$, corresponding to the outcomes under treatment and no treatment, respectively. The observed outcome at time t is $y_t = d_t y_t^1 + (1 - d_t) y_t^0$, where $d_t = \mathbf{1}\{t > T_0\}$.

The fundamental problem of causal inference is that for $t > T_0$, we observe only the treated potential outcome $y_t^1 = y_t$, while the untreated potential outcome y_t^0 remains unobserved. Let the SHC estimator for y_t^0 be denoted $\hat{y}_t^0 := y_t^{\text{SHC}}$ for all $t \in \mathcal{T}_2$. Define the observed post-treatment outcome vector as $\mathbf{y}_{\text{post}} := (y_t^1)_{t \in \mathcal{T}_2}$ and the estimated counterfactual vector as $\mathbf{y}_{\text{post}}^0 := (\hat{y}_t^0)_{t \in \mathcal{T}_2}$. The out-of-sample treatment effect is

$$\alpha_t := y_t^1 - y_t^0, \quad \text{for } t \in \mathcal{T}_2, \quad (2.1)$$

with estimator $\hat{\alpha}_t := y_t - \hat{y}_t^0$. Stacking over $t \in \mathcal{T}_2$ yields:

$$\hat{\boldsymbol{\alpha}} := \mathbf{y}_{\text{post}} - \mathbf{y}_{\text{post}}^0.$$

The average treatment effect on the treated (ATT) is then given by:

$$\widehat{\text{ATT}} := \frac{1}{n} \sum_{t=T_0+1}^{T_0+n} (y_t - \hat{y}_t^0) = \frac{1}{n} \mathbf{1}^\top \hat{\boldsymbol{\alpha}}, \quad (2.2)$$

where $\mathbf{1} \in \mathbb{R}^n$ is a vector of ones.

2.2. The Model

The SHC method introduced by [Chen et al. \(2024\)](#) constructs a counterfactual using the treated unit's own pre-treatment time series.¹ SHC requires a few assumptions:

ASSUMPTION 2.1. (ERROR PROPERTIES AND REGRESSOR INDEPENDENCE) *The error term satisfies $\mathbb{E}[\varepsilon_t] = 0$ and $\mathbb{E}[\varepsilon_t^2] < \infty$ for all $t \in \mathcal{T}$.*

Or the mean of the residuals is 0 in expectation. This fairly standard assumption says any unmodeled fluctuations or noise in the outcome data are purely random and do not systematically bias the estimation of the counterfactual.

ASSUMPTION 2.2. (LATENT COMPONENT SMOOTHNESS AND MATCHING CONDITION) *The latent trend $\tilde{y}(t)$ is $(H + 1)$ times differentiable for some integer $H \geq 3$, with*

¹SCM estimates the counterfactual by solving:

$$\mathbf{w}^* = \underset{\mathbf{w} \in \Delta^{N_0}}{\text{argmin}} \|\mathbf{y}_1 - \mathbf{Y}_0 \mathbf{w}\|_2^2, \quad \forall t \in \mathcal{T}_1, \quad (2.3)$$

where $\mathbf{Y}_0 \in \mathbb{R}^{T_0 \times N_0}$ is the donor matrix of unexposed units and $\Delta^{N_0} = \{\mathbf{w} \in \mathbb{R}_{\geq 0}^{N_0} : \|\mathbf{w}\|_1 = 1\}$ is the probability simplex.

derivative bounded as $|\tilde{y}^{(H+1)}(t)| < \varepsilon$ uniformly over $t \in \mathcal{T}_1$. There exists a convex weight vector $\mathbf{w}^* \in \Delta^{N-1}$ such that the evaluation subvector $\mathbf{y}_{eval}$ can be exactly matched by a convex combination of donor segments: $\mathbf{y}_{eval} = \tilde{\mathbf{Y}}_{pre} \mathbf{w}^*$.

Assumption (a) allows us to approximate the post-treatment latent component using a local linear extrapolation of pre-treatment values. Assumption (b) ensures that a synthetic match exists within the donor pool for the evaluation window, much as [Abadie et al. \(2010\)](#) assume.

ASSUMPTION 2.3. (ESTIMATION FEASIBILITY AND NUMERICAL REGULARITY) *The estimated evaluation window $\mathbf{y}_{eval}$ is exactly matched by the estimated donor matrix: $\hat{\mathbf{y}}_{eval} = \tilde{\mathbf{Y}}_{pre} \mathbf{w}$ for some $\mathbf{w} \in \Delta^{N-1}$. The estimated weights $\mathbf{w}$ have sparse support: the number of non-zero entries is bounded above by a constant $s \ll N$. The submatrix of selected donors $\tilde{\mathbf{Y}}_{pre}^{(s)}$ satisfies that its Gram matrix is positive definite: $(\tilde{\mathbf{Y}}_{pre}^{(s)})^\top \tilde{\mathbf{Y}}_{pre}^{(s)} \succ 0$.*

These sample-level assumptions ensure that the SHC weights are identifiable, sparse, and numerically stable. Much as with SCM, we prefer that a sparse subset of historical controls can approximate the final evaluation period, as is discussed by [Abadie \(2021\)](#).

The first step of the SHC smooths $\mathbf{y}_{pre}$ using local linear regression, where we fit the regression model:

$$y_j \approx a_t + b_t(j - t), \quad \text{for } j = 1, \dots, T_0, \quad (2.4)$$

with local intercept $a_t \in \mathbb{R}$ and slope $b_t \in \mathbb{R}$. Define the centered time vector $\mathbf{s}_t = (1 - t, 2 - t, \dots, T_0 - t)^\top \in \mathbb{R}^{T_0}$, and the design matrix $\mathbf{X}_t = [\mathbf{1} \ \mathbf{s}_t] \in \mathbb{R}^{T_0 \times 2}$. The diagonal kernel weight matrix is:

$$\mathbf{W}_t = \text{diag}(w_{1t}, w_{2t}, \dots, w_{T_0t}), \quad w_{jt} \propto \exp\left(-\frac{1}{2} \left(\frac{j - t}{h}\right)^2\right), \quad (2.5)$$

where $(h > 0)$ is a bandwidth parameter governing smoothness. The local coefficients $\boldsymbol{\beta}_t = (a_t, b_t)^\top$ solve quadratic programming problem that is strictly convex

$$\boldsymbol{\beta}_t = \underset{\boldsymbol{\beta} \in \mathbb{R}^2}{\text{argmin}} \left\| \mathbf{W}_t^{1/2} (\mathbf{y}_{pre} - \mathbf{X}_t \boldsymbol{\beta}) \right\|_2^2. \quad (2.6)$$

Fortunately, this optimization admits a closed-form solution for the coefficients:

$$\boldsymbol{\beta}_t = (\mathbf{X}_t^\top \mathbf{W}_t \mathbf{X}_t)^{-1} \mathbf{X}_t^\top \mathbf{W}_t \mathbf{y}_{pre}. \quad (2.7)$$

The smoothed outcome at time t is $\tilde{y}_t = a_t = \mathbf{e}_1^\top \boldsymbol{\beta}_t$, where $\mathbf{e}_1 = (1, 0)^\top$. Let $\tilde{\mathbf{y}} = (\tilde{y}_1, \dots, \tilde{y}_{T_0})^\top \in \mathbb{R}^{T_0}$ denote the full smoothed series. After we have smoothed over the pre-treatment period before the evaluation period, we build our donor matrix of historical control segments/units.

In the second step, SHC constructs $N = T_0 - m - n + 1$ overlapping temporal segments of length m . Each segment $\tilde{\mathbf{y}}_{[i, i+m-1]}$ is a contiguous subvector of the smoothed pre-treatment series. These comprise the donor matrix $\tilde{\mathbf{Y}}_{pre} := [\tilde{\mathbf{y}}_{[1, m]} \ \tilde{\mathbf{y}}_{[2, m+1]} \ \dots \ \tilde{\mathbf{y}}_{[N, N+m-1]}] \in \mathbb{R}^{m \times N}$.

EXAMPLE 2.1. (CONSTRUCTING $\tilde{\mathbf{Y}}_{\text{PRE}}$) Suppose we have $T_0 = 60$ months of pre-treatment data and we wish to construct donor segments of length $m = 12$ months in order to predict a post-treatment horizon of $n = 6$ months. The number of eligible segments is: $N = T_0 - m - n + 1 = 60 - 12 - 6 + 1 = 43$.

Each donor segment is a vector of 12 consecutive smoothed pre-treatment outcomes.

In the third step, we solve the following program to obtain the optimal weights:

$$\begin{aligned} \mathbf{w}^* = \underset{\mathbf{w} \in \mathbb{R}^N}{\text{argmin}} \quad & \left\| \tilde{\mathbf{y}}_{\text{eval}} - \tilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \\ \text{subject to} \quad & \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1 \end{aligned} \quad (2.8)$$

Chen et al. (2024) note that in the general case when we have more donor units than we do the length of the segments ($N > m$), the optimization becomes an ill-posed problem. The regularization term stabilizes the optimization by penalizing weight allocations along directions of low variance in the donor pool, reducing sensitivity to collinear segments. Formally, $\mathbf{C}_0$ contains the eigenvectors of the Grammian matrix $\mathbf{G} \in \mathbb{R}^{N \times N}$. $\mathbf{C}_0$ corresponds to the eigenvalues less than $\tau > 0$. These directions capture low-variance combinations of donor segments, and the penalty shrinks weights in those directions.² The corresponding weight vector $\mathbf{w}^*$ is used to compute the counterfactual by applying it to the shifted donor segments:

$$\tilde{\mathbf{Y}}_{\text{post}} = [\tilde{\mathbf{y}}_{[1+m, m+n]} \quad \tilde{\mathbf{y}}_{[2+m, m+n+1]} \quad \cdots \quad \tilde{\mathbf{y}}_{[N+m, N+m+n-1]}] \in \mathbb{R}^{n \times N}. \quad (2.9)$$

This yields in-sample and out-of-sample predictions for the SHC estimator. The full counterfactual vector is then: $\mathbf{y}^0 = \tilde{\mathbf{Y}} \mathbf{w}^* \in \mathbb{R}^{m+n}$.

Bandwidth Selection via Cross-Validation To select the bandwidth parameter h used in local linear smoothing of $\mathbf{y}_{\text{pre}}$, I implement a *leave-one-out cross-validation* (LOOCV) procedure over a grid of candidate values $h \in [h_{\min}, h_{\max}] \subset \mathbb{R}_{>0}$. This procedure chooses the bandwidth that minimizes out-of-sample prediction error for the smoothed pre-treatment series $\tilde{\mathbf{y}} \in \mathbb{R}^{T_0}$, as defined in Equation~(2.7).

For each candidate h , and for each time point $t \in \mathcal{T}_1$, define Gaussian kernel weights

$$w_{\tau t}^{(h)} = \exp \left(-\frac{1}{2} \left(\frac{\tau - t}{h} \right)^2 \right), \quad \tau \in \mathcal{T}_1 \setminus \{t\},$$

used to construct the diagonal matrix $\mathbf{W}_t^{(-t)} \in \mathbb{R}^{(T_0-1) \times (T_0-1)}$, omitting the index t . Let $\mathbf{X}_t^{(-t)} \in \mathbb{R}^{(T_0-1) \times 2}$ and $\mathbf{y}_{\text{pre}}^{(-t)} \in \mathbb{R}^{T_0-1}$ denote the design matrix and response vector with time t excluded.

Then the local linear coefficients $\beta_t^{(-t)} = (a_t^{(-t)}, b_t^{(-t)})^\top$ are obtained by solving Equation~(2.6). The leave-one-out smoothed value at time t is then:

$$\tilde{y}_t^{(-t)} = a_t^{(-t)} = \mathbf{e}_1^\top \beta_t^{(-t)}, \quad \text{where } \mathbf{e}_1 = (1, 0)^\top.$$

²The ς (sigma) term is a small positive constant used to stabilize the solution in cases of near multicollinearity.

To pick the ideal bandwidth, we minimize:

$$h^* = \underset{h \in [h_{\min}, h_{\max}]}{\operatorname{argmin}} \left\{ \operatorname{CV}(h) := \frac{1}{T_0} \sum_{t=1}^{T_0} \left(y_t - \hat{y}_t^{(-t)} \right)^2 \right\}. \quad (2.10)$$

Once selected, the optimal bandwidth h^* is used to construct the full smoothed pre-treatment series.

Control Group Selection [Chen et al. \(2024\)](#) note that we should have a parsimonious approach to which historical controls we use, rather than choosing all of them. In their examples, SHC typically requires less than 28 controls. They advocate for a forward selection procedure to choose the optimal control group. I implement a slightly modified variant of this method to choose the ideal number of historical controls. Let $\mathcal{N} = \{1, 2, \dots, N\}$ index the full set of donor segments in the pre-treatment matrix $\tilde{\mathbf{Y}}_{\text{pre}} \in \mathbb{R}^{m \times N}$. At each step the algorithm selects the donor index $i_j \in \mathcal{N} \setminus \mathcal{S}_{j-1}$ that, when added to the current donor set $\mathcal{S}_{j-1}$, minimizes the in-sample mean squared error (MSE) between the evaluation window and in-sample fit:

$$\mathcal{S}_j = \mathcal{S}_{j-1} \cup \left\{ \underset{i \in \mathcal{N} \setminus \mathcal{S}_{j-1}}{\operatorname{argmin}} \left\| \tilde{\mathbf{y}}_{\text{eval}} - \tilde{\mathbf{Y}}_{\text{pre}}^{(\mathcal{S}_{j-1} \cup \{i\})} \mathbf{w}^{(j)} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \right\}, \quad \mathcal{S}_0 = \emptyset,$$

where $\tilde{\mathbf{Y}}_{\text{pre}}^{(\mathcal{S})}$ denotes the submatrix of $\tilde{\mathbf{Y}}_{\text{pre}}$ with columns indexed by $\mathcal{S}$, and the weights $\mathbf{w}^{(j)} \in \Delta^j$ are the optimal simplex-constrained weights for the selected donor set $\mathcal{S}_j$. This procedure greedily expands the donor set one segment at a time, always choosing the segment that yields the greatest marginal reduction in in-sample error.

To avoid overfitting, [Chen et al. \(2024\)](#) manually choose the number of donors the forward selection procedure should choose. In their case, they experiment with the single best historical segment (which rarely does well), the best 15, 20, and 27 historical control units. However, this is a subjective decision that will yield different results depending on what we select. To mitigate researcher degrees of freedom, I view the number of historical segments as a tuning parameter. I employ a forward selection-based algorithm to choose the number of controls. For my implementation of SHC, I employ an early stopping rule based on a modified Bayesian information criterion (BIC), borrowing inspiration from [Shi and Huang \(2023\)](#). For each donor set size j , I compute:

$$\operatorname{BIC}(j) = m \cdot \log(\operatorname{MSE}_j) + \lambda j,$$

where MSE_j is the in-sample error using donor set $\mathcal{S}_j$, and $\lambda = \log(m)$ penalizes the inclusion of too many donors. The selection procedure terminates when $\operatorname{BIC}(j)$ increases for two consecutive steps. The final donor set $\mathcal{S}^*$ is then taken to be $\mathcal{S}_{j^*}$, where:

$$j^* = \underset{j}{\operatorname{argmin}} \operatorname{BIC}(j).$$

This approach adaptively selects a sparse donor subset that closely matches the treated

unit's pre-intervention trajectory. The selected donor set $\mathcal{S}^*$ is used to compute the SHC model predictions. I would like to mention that this forward selection method holds much in common with recently proposed methods from the literature. For example, [Shi and Huang \(2023\)](#) propose the forward selected panel data approach which selects the comparison group for the panel data approach by [Hsiao et al. \(2012\)](#) using an aggressive forward selection procedure. [Li \(2024\)](#), correcting for overfitting in the former, proposed the forward Difference-in-Differences estimator, where a forward selection approach is used to select the donor pool for the Difference-in-Differences model. Similar advances have been made for the synthetic control method, as developed by [Cerulli \(2024\)](#), where a similar selection mechanism is used to estimate the treatment effect. It would be interesting, but is far outside the scope of this work, to compare these estimators to one another and see when their in-sample fits and point estimates agree with one another.

Inference Quantifying uncertainty around the counterfactual predictions is essential for valid inference. [Chen et al. \(2024\)](#) do not provide any inferential method to use aside from in-time placebos. Following recent work in post-modeling inference by [Cattaneo et al. \(2025\)](#), I construct agnostic conformal prediction intervals around the counterfactual series. These intervals are distribution-free, require no parametric assumptions, and are valid under exchangeability of residuals between the pre- and post-treatment periods (or, good in-sample fit).

Let $\mathbf{y}_{\text{eval}}^0 \in \mathbb{R}^m$ denote the SHC-fitted values during the evaluation period and define the residuals as:

$$\boldsymbol{\varepsilon}_{\text{eval}} := \mathbf{y}_{\text{eval}} - \mathbf{y}_{\text{eval}}^0.$$

I then compute the sample mean $\bar{\varepsilon}$ and variance $\hat{\sigma}^2$ of these residuals:

$$\bar{\varepsilon} = m^{-1} \sum_{t \in \mathcal{T}_{\text{eval}}} \varepsilon_t, \quad \hat{\sigma}^2 = (m-1)^{-1} \sum_{t \in \mathcal{T}_{\text{eval}}} (\varepsilon_t - \bar{\varepsilon})^2.$$

To construct prediction intervals for the post-treatment counterfactual $\mathbf{y}_{\text{post}}^0 \in \mathbb{R}^n$, I assume the residuals are sub-Gaussian and apply a conformal bound based on Hoeffding-type concentration inequalities. For a miscoverage rate $\alpha \in (0, 1)$, the interval half-width is given by:

$$c_\alpha = \sqrt{2\hat{\sigma}^2 \log\left(\frac{2}{\alpha}\right)}.$$

Then, the agnostic $(1 - \alpha)$ prediction interval for each $t \in \mathcal{T}_2$ is given by:

$$[\hat{y}_t^0 + \bar{\varepsilon} - c_\alpha, \hat{y}_t^0 + \bar{\varepsilon} + c_\alpha].$$

Stacking these yields lower and upper bound vectors:

$$\mathbf{L} := \mathbf{y}_{\text{post}}^0 + \bar{\varepsilon} \cdot \mathbf{1} - c_\alpha \cdot \mathbf{1}, \quad \mathbf{U} := \mathbf{y}_{\text{post}}^0 + \bar{\varepsilon} \cdot \mathbf{1} + c_\alpha \cdot \mathbf{1},$$

which define the conformal confidence band around the counterfactual trajectory.

This approach allows finite-sample valid coverage under the assumption that the pre- and post-treatment residuals are exchangeable. Intuitively, if the SHC model fits the pre-treatment series well (i.e., residuals are small and centered), then its extrapolations into the post-treatment period are expected to carry similarly bounded uncertainty. In this sense, the residual distribution learned from the evaluation window is used to calibrate uncertainty for the out-of-sample outcomes.

These prediction intervals reflect uncertainty in the estimated counterfactual. Hence, if the observed outcome falls far outside the prediction interval, this is interpreted as evidence of a treatment effect. Unlike traditional confidence intervals around treatment effect estimates (e.g., using standard errors), the conformal intervals here do not assume a data-generating model. I use a default coverage level of 90% (i.e., $\alpha = 0.1$) in all applications.

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